Do Now · retrieval

byVal, byRef and modular programming

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does return do in a C# function?

Q2

Can a method read a local variable declared inside another method?

Q3

What is a[0] for int[] a={9,6,3}?

Q4

If Double(n) returns 2*n, find Double(Double(3)).

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

CALLER — The caller declares marks = 10.

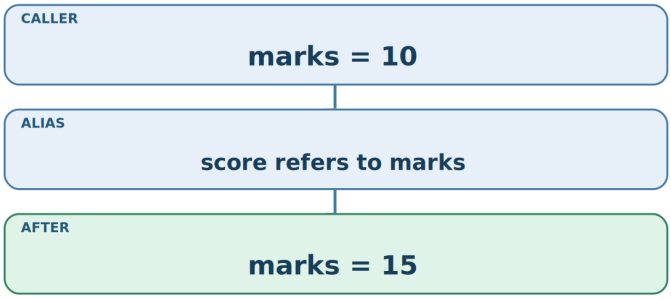
Core idea 2

ALIAS — ref makes score refer to the caller's marks rather than an independent copy.

Core idea 3

AFTER — Changing score therefore changes marks, which becomes 15.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

by value — The subroutine receives a copy of the value.

by reference — The parameter refers to the caller's variable.

modularity — Divide a program into smaller testable subroutines.

ref — C# keyword enabling reference parameter passing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

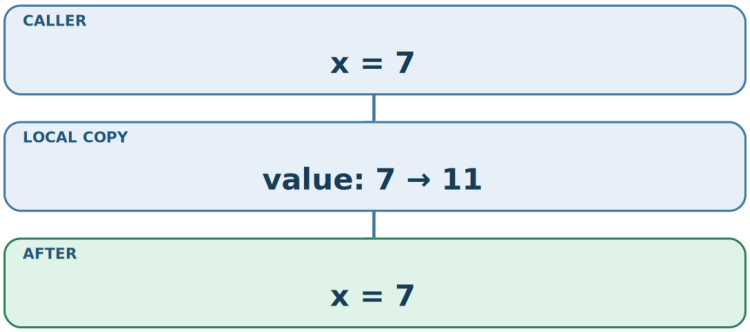
Problem

A function receives a copy of x=7 and adds 4. What is x afterwards?

Follow the worked method

1. Predict first — A new parameter value receives a copy of the integer 7.
2. Model the result — value becomes 11 inside AddFour, but x remains 7 after the call.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
y starts at 8 and is passed using ref.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A method takes int number by value and adds 2. Caller x=5. What is x after the call?

Question 2

Write a C# procedure that doubles an integer passed by reference.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain when returning a result is preferable to changing a ref parameter.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Changing a by-value int parameter always changes the caller's int.

Use key vocabulary from memory

1. Explain by value in your own words.

2. Explain by reference in your own words.

3. Explain modularity in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.